Energy-Efficient Photonic Neurons via High-Q Microring Resonators: Theoretical Foundations, Design Constraints, and Bandwidth Tradeoffs

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Abstract

Photonic neurons offer a pathway to ultrafast, low-energy neuromorphic computation by exploiting the optical Kerr nonlinearity in integrated waveguides to implement activation-like nonlinear transfer functions entirely in the optical domain. A central challenge is achieving the π-radian phase shift required for a meaningful nonlinear response at practical input power levels. High-quality-factor (Q) microring resonators address this challenge through two coupled mechanisms: resonant field enhancement, which amplifies the intracavity intensity beyond the input level, and resonant buildup of effective interaction length, which increases the nonlinear phase accumulation per unit of input power. Together these scale the required input energy as $E \propto 1/Q^2$, enabling femtojoule-level switching. This gain comes at a fundamental cost: photon lifetime $\tau_{\mathrm{ph}} \propto Q$ broadens and slows the impulse response, constraining the operational bandwidth to $\Delta\nu \propto 1/Q$. This paper develops the theoretical framework grounding these mechanisms in Agrawal's nonlinear Schrödinger (NLS) formalism [1], presents a literature-supported silicon nitride microring design, analyzes placement and coupling constraints for optical computing, and evaluates the bandwidth–energy tradeoff in detail. Future work aims to recover high-speed operation while retaining low-intensity operation, through structural modifications or optimized tradeoff analysis.

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1 Introduction and Motivation

Optical computing has long been attractive due to the potential for operating at the speed of light with minimal heat dissipation. A photonic neuron implements a nonlinear activation function entirely in the optical domain, avoiding the latency and energy overhead of optical-to-electronic conversion. The core physical mechanism is the optical Kerr effect: an intensity-dependent refractive index of the form

$$\tilde{n}(\omega, I) = n(\omega) + n_2 I, \tag{1}$$

where $n(\omega)$ is the linear refractive index, I is the local optical intensity, and n_2 is the nonlinear-index (Kerr) coefficient [1]. As Agrawal shows in Chapter 1.3 (Eq. 1.3.2), the intensity dependence arises from the real part of the third-order susceptibility $\chi^{(3)}$:

$$n_2 = \frac{3}{8n} \operatorname{Re}\left(\chi^{(3)}_{xxxx}\right). \tag{2}$$

For silica, $n_2 \approx 2.2$–3.4×10^{-20} m^2/W [1]. Silicon nitride (Si_3N_4), the material of the proposed device, has a substantially higher $n_2 \approx 2.4$–6.9×10^{-19} m^2/W, offering an order-of-magnitude enhancement in nonlinear response at comparable loss [3, 4].

The fundamental challenge is that achieving a π-radian Kerr phase shift—sufficient for full-extinction switching in an interferometric geometry—requires intensities that are impractically large in a straight waveguide of typical chip-scale length ($\lesssim 1$ mm). High-Q microring resonators solve this problem by resonantly recycling the optical field, concentrating it in a compact volume and multiplying both the effective intensity and the effective interaction length. The following sections develop this argument rigorously, starting from the NLS propagation equation.

2 Propagation Framework: The Nonlinear Schrödinger Equation

The evolution of a slowly varying pulse envelope $A(z,t)$ in a single-mode waveguide is governed by the generalized nonlinear Schrödinger (NLS) equation. In its standard form (Agrawal, Eq. 2.3.28 [1]):

$$\frac{\partial A}{\partial z} + \beta_1 \frac{\partial A}{\partial t} + \frac{i\beta_2}{2}\frac{\partial^2 A}{\partial t^2} + \frac{\alpha}{2} A = i\gamma |A|^2 A, \tag{3}$$

where $\beta_1 = 1/v_g$ is the inverse group velocity, β_2 is the group-velocity dispersion (GVD) parameter, α is the propagation loss coefficient, and γ is the nonlinear parameter defined by Agrawal (Eq. 2.3.30):

$$\gamma = \frac{\omega_0 n_2}{c A_{\text{eff}}}, \tag{4}$$

with A_{eff} the effective mode area (Agrawal, Eq. 2.3.31):

$$A_{\text{eff}} = \frac{\left(\iint_{-\infty}^{\infty} |F(x,y)|^2 \, dx\, dy\right)^2}{\iint_{-\infty}^{\infty} |F(x,y)|^4 \, dx\, dy}, \tag{5}$$

where $F(x,y)$ is the transverse modal distribution. The parameter γ therefore encodes the interplay between material nonlinearity (n_2), operating frequency (ω_0), and modal confinement (A_{eff}): tighter confinement reduces A_{eff} and raises γ, directly reducing the power threshold for nonlinear effects.

2.1 SPM Phase Shift and Nonlinear Length

When dispersion effects are negligible (i.e., over lengths $L \ll L_D = T_0^2/|\beta_2|$), Eq. (3) reduces to the pure SPM equation (Agrawal, Eq. 4.1.1):

$$\frac{\partial A}{\partial z} = \frac{i}{L_{\text{NL}}} |A|^2 A, \tag{6}$$

where the nonlinear length is (Agrawal, Eq. 4.1.2):

$$L_{\mathrm{NL}} = (\gamma P_0)^{-1}. \tag{7}$$

The solution gives a purely phase-modulated output (Agrawal, Eqs. 4.1.4–4.1.7):

$$A(L,T) = A(0,T)\exp[i\phi_{\mathrm{NL}}(L,T)]\,, \qquad \phi_{\mathrm{NL}}(L,T) = |A(0,T)|^2 \frac{L_{\mathrm{eff}}}{L_{\mathrm{NL}}}, \tag{8}$$

with effective length $L_{\mathrm{eff}} = [1-e^{-\alpha L}]/\alpha$. The maximum nonlinear phase shift at the pulse center is therefore:

$$\phi_{\max} = \gamma P_0 L_{\mathrm{eff}}. \tag{9}$$

Setting $\phi_{\max} = \pi$ and solving for input power:

$$P_{\pi,\,\mathrm{straight}} = \frac{\pi}{\gamma L_{\mathrm{eff}}} = \frac{\pi c A_{\mathrm{eff}}}{\omega_0 n_2 L_{\mathrm{eff}}}. \tag{10}$$

For a Si_3N_4 waveguide with $n_2 = 2.4 \times 10^{-19}$ m^2/W, $A_{\mathrm{eff}} = 1$ μm^2, and $L = 1$ mm, one finds $P_{\pi,\,\mathrm{straight}} \sim$ several watts—far beyond what is suitable for dense, low-power neural arrays. The role of the microring resonator is to reduce this threshold by orders of magnitude.

2.2 Nonlinear Transfer Function for a Microring-Based Photonic Neuron

For a resonator-embedded neuron, the relevant transfer function is the all-pass or add-drop microring transmission, which has a Lorentzian (resonance) profile:

$$T(\Delta\omega) = \frac{(\Delta\omega/2)^2}{(\Delta\omega/2)^2 + (\omega-\omega_r)^2}, \tag{11}$$

where ω_r is the resonance frequency and $\Delta\omega = \omega_r/Q$ is the linewidth. Near resonance, the intensity-dependent shift of ω_r via the Kerr effect produces a strongly nonlinear input-output relationship. This resonator transfer function, combined with the Kerr phase shift, forms the optical analogue of a sigmoidal activation function: it is approximately linear for small input powers and saturates (or switches) at higher powers, providing the threshold-like behavior required for neural computation.

3 Why a Microring Resonator? Resonant Enhancement of Kerr Nonlinearity

The fundamental motivation for embedding a Kerr-active waveguide in a ring resonator is resonant field enhancement: the intracavity field amplitude builds up coherently over many round trips when the input is on resonance, concentrating optical energy in a small volume and dramatically reducing the external power needed to achieve a given nonlinear phase shift.

3.1 Intracavity Intensity Enhancement

For a lossless resonator with power transmission coefficient t^2 at the coupling point, the intracavity power at resonance exceeds the input power by the buildup factor [2]:

$$B = \frac{P_{\mathrm{cav}}}{P_{\mathrm{in}}} = \frac{1-r^2}{(1-ra)^2}, \tag{12}$$

where r is the self-coupling coefficient and a is the round-trip amplitude transmission. Under critical coupling ($r = a$, the condition that maximizes power transfer into the ring), the input power is fully absorbed by internal losses and Eq. (12) simplifies to:

$$B\big|_{\mathrm{crit}} = \frac{1}{1-r^2} \approx \frac{\mathcal{F}}{\pi}, \tag{13}$$

where $\mathcal{F} = \pi\sqrt{ra}/(1-ra)$ is the cavity finesse [2]. The finesse is directly proportional to the quality factor:

$$\mathcal{F} = \frac{\lambda}{n_g L} \cdot Q, \tag{14}$$

so the intracavity intensity enhancement scales linearly with Q:

$$I_{\text{cav}} \propto Q \cdot I_{\text{in}}. \tag{15}$$

3.2 Resonantly Enhanced Effective Interaction Length

A photon circulating in a resonator does not traverse the nonlinear medium just once but makes many round trips before leaking out. The mean number of round trips before photon escape is proportional to Q, so the effective interaction length seen by the stored photons is:

$$L_{\text{eff, res}} = \frac{Q\lambda}{2\pi n_g}, \tag{16}$$

where n_g is the group index of the waveguide material. This replaces the geometric ring circumference L as the relevant propagation length for nonlinear phase accumulation. The photon-lifetime-based effective length can significantly exceed the physical device dimensions: for $Q = 10^6$, $\lambda = 1550$ nm, and $n_g = 2.0$, one finds $L_{\text{eff, res}} \approx 12$ cm, more than three orders of magnitude larger than a ~ 100 μm ring circumference.

3.3 Q^2 Energy Scaling

Substituting both resonant enhancements—intensity (Eq. 15) and effective length (Eq. 16)—into the π-phase condition $\Delta\phi = (2\pi/\lambda) n_2 I_{\text{cav}} L_{\text{eff, res}}$ and solving for the external input intensity:

$$I_{\text{in}} \propto \frac{n_g}{n_2 Q^2}\,\Delta\phi, \tag{17}$$

so the required input pulse energy scales as:

$$\boxed{E = I_{\text{in}}\, A_{\text{eff}}\, \tau \propto \frac{n_g\, A_{\text{eff}}\, \tau}{n_2\, Q^2}\,\Delta\phi.} \tag{18}$$

This is the central result: the operating energy decreases as the *square* of the quality factor. Doubling Q reduces the required energy by a factor of four. This quadratic enhancement arises because Q simultaneously improves both the intracavity intensity (one factor of Q from field buildup) and the effective path length (one factor of Q from photon lifetime), and both reductions compound multiplicatively in the phase integral.

This result is consistent with the enhancement of nonlinear efficiency described by Agrawal (Eqs. 1.3.10–1.3.12), where for a straight fiber the nonlinear figure of merit $I_0 L_{\text{eff}}$ increases with decreasing mode area; the resonator provides an additional Q^2 boost by recycling the field.

4 Microring Design Parameters: Literature-Supported Values

The choice of platform, geometry, and coupling parameters is critical to realizing the theoretical Q^2 benefit in practice. This section presents a detailed, literature-grounded design for a Si_3N_4 microring resonator optimized for photonic neural operation at 1550 nm.

4.1 Material Platform: Silicon Nitride

Silicon nitride is the platform of choice for this application for several compounding reasons:

- **High Kerr nonlinearity.** The nonlinear-index coefficient of Si_3N_4 is $n_2 \approx 2.4$–6.9×10^{-19} m^2/W [3, 4], roughly 10× larger than silica, directly reducing the required operating power via Eq. (18).

- **Ultra-low propagation loss.** State-of-the-art Si_3N_4 waveguides achieve propagation losses below 1 dB/m (i.e., < 0.1 dB/cm) [4, 5], preserving the high intracavity field that drives Kerr switching.
- **Wide transparency window.** Si_3N_4 is transparent from $\sim$400 nm to $>$2.3 μm [3], encompassing the telecom C-band (1530–1565 nm) targeted here.
- **No two-photon absorption (TPA) at 1550 nm.** Unlike silicon, Si_3N_4 has a bandgap ($\sim$ 5 eV) wide enough to avoid TPA at telecom wavelengths, which would otherwise clamp the intracavity power and limit Q [6].
- **CMOS compatibility.** Si_3N_4 is deposited by low-pressure chemical vapor deposition (LPCVD) and is fully compatible with CMOS back-end-of-line processing, supporting integration with electronic control and readout circuitry [4].

4.2 Waveguide Cross-Section and Mode Area

For the fundamental quasi-TE mode at 1550 nm, a Si_3N_4 waveguide core of 800×400 nm (width $\times$ height) on SiO_2 cladding confines the mode to an effective area of $A_{\text{eff}} \approx 0.9$–$1.1$ μm^2 [4, 5]. This tight confinement raises γ via Eq. (4):

$$\gamma = \frac{\omega_0 n_2}{c A_{\text{eff}}} \approx \frac{(1.21 \times 10^{15})(2.4 \times 10^{-19})}{(3 \times 10^8)(1 \times 10^{-12})} \approx 0.97 \text{ W}^{-1}\text{m}^{-1} \approx 1 \text{ W}^{-1}\text{m}^{-1}, \tag{19}$$

consistent with reported experimental values of $\gamma = 1$–3 $\text{W}^{-1}\text{m}^{-1}$ for this platform [4]. The group index for this cross-section at 1550 nm is $n_g \approx 1.98$–2.05, obtained from finite-element mode simulations [5].

4.3 Ring Radius, Circumference, and Free Spectral Range

The ring radius R sets both the free spectral range (FSR) and the bending-loss contribution. For $R = 23$ μm, the circumference is:

$$L = 2\pi R \approx 144 \ \mu\text{m}. \tag{20}$$

The FSR is:

$$\text{FSR} = \frac{c}{n_g L} \approx \frac{3 \times 10^8}{2.0 \times 144 \times 10^{-6}} \approx 1.04 \text{ THz}, \tag{21}$$

or equivalently $\sim$ 8.3 nm at 1550 nm in wavelength units. A large FSR is essential for optical computing to avoid spurious resonances overlapping with adjacent signal or pump wavelengths. Bending loss for this radius is negligible (< 0.01 dB/cm) for this waveguide cross-section [4].

4.4 Coupling Gap and Critical Coupling Condition

The coupling gap g between the bus waveguide and the ring determines the power coupling coefficient $\kappa^2 = 1 - r^2$. For critical coupling, the coupling loss must equal the round-trip loss:

$$\kappa^2 = 1 - a^2 = \alpha_{\text{prop}} \cdot L, \tag{22}$$

where α_{prop} is the power propagation loss per unit length. For $\alpha_{\text{prop}} \approx 1$ dB/m $\approx 2.3 \times 10^{-4}$ m^{-1} and $L = 144$ μm:

$$\kappa^2 \approx (2.3 \times 10^{-4})(144 \times 10^{-6}) \approx 3.3 \times 10^{-8}. \tag{23}$$

This extremely small coupling coefficient corresponds to a coupling gap of $g \approx 200$–300 nm for this waveguide geometry [2, 5]. Achieving such gaps reliably requires deep-UV or e-beam lithography, which is available at leading photonic foundries.

4.5 Quality Factor

The intrinsic quality factor Q_i is set by propagation loss, while the loaded Q_L accounts for coupling loss. At critical coupling, $Q_L = Q_i/2$. The intrinsic Q is:

$$Q_i = \frac{2\pi n_g}{\lambda \alpha_{\text{prop}}} \approx \frac{2\pi(2.0)}{(1550 \times 10^{-9})(2.3 \times 10^{-4})} \approx 3.5 \times 10^7. \tag{24}$$

In practice, surface roughness and fabrication imperfections reduce this. State-of-the-art fabricated Si_3N_4 rings have achieved loaded Q_L values of 1×10^7 [5] and intrinsic values exceeding 6.7×10^7 [4]. For this proposal, a conservative target of $Q = 10^6$–10^7 is adopted.

4.6 Photon Lifetime

The photon lifetime τ_{ph} is determined by the loaded Q:

$$\tau_{\text{ph}} = \frac{Q\lambda}{2\pi c} = \frac{Q}{\omega_0/Q} = \frac{Q}{\Delta\omega}. \tag{25}$$

For $Q = 10^6$ and $\lambda = 1550$ nm:

$$\tau_{\text{ph}} = \frac{10^6 \times 1550 \times 10^{-9}}{2\pi \times 3 \times 10^8} \approx 0.82 \text{ ns}. \tag{26}$$

For $Q = 10^7$, $\tau_{\text{ph}} \approx 8.2$ ns. This will be critical to the bandwidth analysis in Section 7.

4.7 Summary of Design Parameters

Parameter	**Value**	**Source**
Wavelength λ	1550 nm	C-band telecom [1]
Material	Si_3N_4 / SiO_2	[3, 4]
Waveguide cross-section	800×400 nm	[4, 5]
n_2	2.4–6.9×10^{-19} m^2/W	[3]
A_{eff}	≈ 1 μm^2	[4]
Nonlinear parameter γ	≈ 1 W^{-1}m^{-1}	[4]
Group index n_g	≈ 2.0	[5]
Ring radius R	23 μm	This work
Circumference L	≈ 144 μm	Calculated
FSR	≈ 1 THz (≈ 8 nm)	Calculated
Coupling gap g	200–300 nm	[2, 5]
Target Q_L	10^6–10^7	[5, 4]
Photon lifetime τ_{ph}	0.8–8 ns	Calculated
Propagation loss	< 1 dB/m	[4]

5 Numerical Estimate of Operating Energy

Using Eq. (18) with the design parameters above, the required input pulse energy to achieve a π-phase shift is:

$$E = \frac{n_g A_{\text{eff}}}{n_2 Q^2} \Delta\phi \cdot \tau. \tag{27}$$

For $Q = 10^6$, $\tau = \tau_{\text{ph}} \approx 0.82$ ns, and $\Delta\phi = \pi$:

$$E \approx \frac{2.0 \times 1 \times 10^{-12}}{2.4 \times 10^{-19} \times 10^{12}} \times \pi \times 0.82 \times 10^{-9} \approx 22 \text{ fJ}. \tag{28}$$

For $Q = 10^7$, the same calculation yields $E \approx 0.2$ fJ. These values are consistent with the femtojoule-scale operation demonstrated in photonic neural network demonstrations [7, 8] and represent improvements of 5–6 orders of magnitude over the straight-waveguide estimate from Eq. (10).

6 Design Constraints and Placement in an Optical Computing Architecture

Realizing the benefits of the microring neuron in a practical optical computing system imposes a set of interconnected design constraints on the device and its placement within the photonic integrated circuit (PIC).

6.1 Thermal Stability and Resonance Control

The resonance wavelength of a Si_3N_4 ring shifts at a rate of approximately $\partial\lambda_r/\partial T \approx 25$–30 pm/K due to the thermo-optic effect ($\partial n/\partial T \approx 2.4 \times 10^{-5}$ K^{-1} for Si_3N_4) [4]. For a resonance linewidth of $\Delta\lambda = \lambda/Q \approx$ 1.55 pm at $Q = 10^6$, a temperature fluctuation of even ∼0.1 K is sufficient to detune the resonance by a full linewidth. This means that practical deployment requires either active thermal stabilization (integrated heaters with feedback) or athermal design using materials with compensating thermo-optic and thermo-mechanical responses. Placing resonators in a thermally isolated region of the PIC, or on a substrate with low thermal conductivity, can relax this requirement.

6.2 Nonlinear Resonance Shift and Bistability

At high intracavity powers, the Kerr effect itself shifts the resonance frequency by $\delta\omega_r \propto n_2 P_{\text{cav}}/A_{\text{eff}}$. If this shift becomes comparable to the resonance linewidth $\Delta\omega = \omega_0/Q$, the resonance becomes a function of input power and the device enters an optical bistable regime [6, 2]. Bistability can be exploited for all-optical memory and latching behavior, but it complicates operation as a simple linear-threshold neuron. The condition for bistability onset is approximately:

$$\frac{2\pi}{\lambda} n_2 \cdot B \cdot \frac{P_\pi}{A_{\text{eff}}} \cdot L > \frac{\Delta\omega}{\omega_0} = \frac{1}{Q}, \tag{29}$$

which at $Q = 10^6$ and the parameters above is approached near P_π. The design must therefore either operate slightly below the bistability threshold or incorporate a stabilizing feedback mechanism.

6.3 Placement: Bus Waveguide Routing and Fanout

In an optical neural network, each photonic neuron receives a weighted sum of optical signals from upstream neurons and emits an output to downstream neurons. The microring's placement must satisfy:

1. **Bus coupling:** The input signal is routed via a straight bus waveguide that evanescently couples into the ring. The coupling gap must be lithographically controlled to within ±10 nm to maintain the critical coupling condition [2].

2. **Drop-port routing:** For add-drop configurations, a second bus waveguide on the opposite side of the ring extracts the processed signal. The drop-port provides complementary transmission to the through-port and can serve as the neuron's output channel.

3. **Weighting network:** Upstream to the ring, a Mach-Zehnder interferometer (MZI) mesh or a microring-based weight bank implements the synaptic weighting of the input signals before they are summed on the input bus [7]. The microring neuron sits at the output node of this mesh.

4. **Fan-out and amplification:** The output of the neuron must drive multiple downstream neurons. On-chip amplification (e.g., semiconductor optical amplifiers or Raman gain sections) may be required to restore signal levels after the inherent insertion loss of the resonator.

5. **Wavelength-division multiplexing (WDM):** Multiple neurons can be parallelized by assigning each a distinct resonance wavelength within the FSR of the ring array. For an FSR of ∼ 1 THz and a linewidth of ∼200 MHz, up to thousands of channels can coexist in principle on a single bus, though practical crosstalk and thermal drift limit this to tens of channels per bus [2].

6.4 Fabrication Tolerances

The performance of the microring is sensitive to fabrication imperfections. Line-edge roughness of the waveguide sidewalls couples the forward-propagating mode to backscattering modes, splitting the resonance into two doublets and reducing the effective Q. For a target $Q = 10^6$–10^7 to be maintained, sidewall roughness must be kept below ∼0.5 nm RMS, achievable with optimized LPCVD deposition and hydrogen annealing [5].

7 Drawbacks: Bandwidth–Quality Factor Tradeoff

The $1/Q^2$ energy scaling derived in Section 3 comes with a fundamental and unavoidable cost: the resonator linewidth $\Delta\nu$ and the photon lifetime τ_{ph} both scale inversely with Q, imposing a hard limit on the speed at which the neuron can respond to changing inputs.

7.1 Resonance Linewidth and Operational Bandwidth

The loaded quality factor is defined as:

$$Q = \frac{\omega_r}{\Delta\omega} = \frac{\nu_r}{\Delta\nu}, \tag{30}$$

so the resonance linewidth (FWHM) in frequency units is:

$$\Delta\nu = \frac{\nu_r}{Q} = \frac{c}{\lambda Q}. \tag{31}$$

The linewidth sets the maximum bandwidth of an optical signal that can efficiently couple into the resonator. Any signal whose spectral content exceeds $\Delta\nu$ will not be fully captured, reducing the effective intracavity power and degrading the Kerr enhancement. Equivalently, the maximum bit rate $B_{\max}$ that the neuron can process is limited by:

$$B_{\max} \lesssim \Delta\nu = \frac{c}{\lambda Q}. \tag{32}$$

Q	$\Delta\nu$ (MHz)	$B_{\max}$ (Gbps)
10^4	19,400	19.4
10^5	1,940	1.9
10^6	194	0.19
10^7	19.4	0.019

The entries in the table above reveal a stark tradeoff: increasing Q from 10^4 to 10^7 reduces the operating energy by $10^6\times$ (via $E \propto 1/Q^2$) but simultaneously reduces the bandwidth by $10^3\times$. At $Q = 10^6$, the device is limited to sub-Gbps operation—too slow for high-speed optical neural network inference, which targets ∼10–100 Gbps operation [7, 8].

7.2 Photon Lifetime and Impulse Response

The photon lifetime $\tau_{\text{ph}} = Q\lambda/(2\pi c)$ is the $1/e$ time constant for energy to escape the resonator. It determines the temporal response of the device to a sudden change in input:

$$\tau_{\text{ph}} = \frac{Q\lambda}{2\pi c} \propto Q. \tag{33}$$

An input pulse shorter than τ_{ph} cannot couple its full energy into the cavity (mismatch between pulse bandwidth and resonance linewidth). An input that switches on suddenly will cause the intracavity energy—and hence the Kerr phase shift—to rise with the time constant τ_{ph}. The output of the neuron therefore rises and falls on the scale of τ_{ph}, imposing the same limit as Eq. (32):

$$\text{Speed} \propto 1/\tau_{\text{ph}} \propto 1/Q. \tag{34}$$

7.3 Energy–Speed Product

Combining Eqs. (18) and (33):

$$E \cdot \tau_{\text{ph}} \propto \frac{1}{Q^2} \cdot Q = \frac{1}{Q}. \tag{35}$$

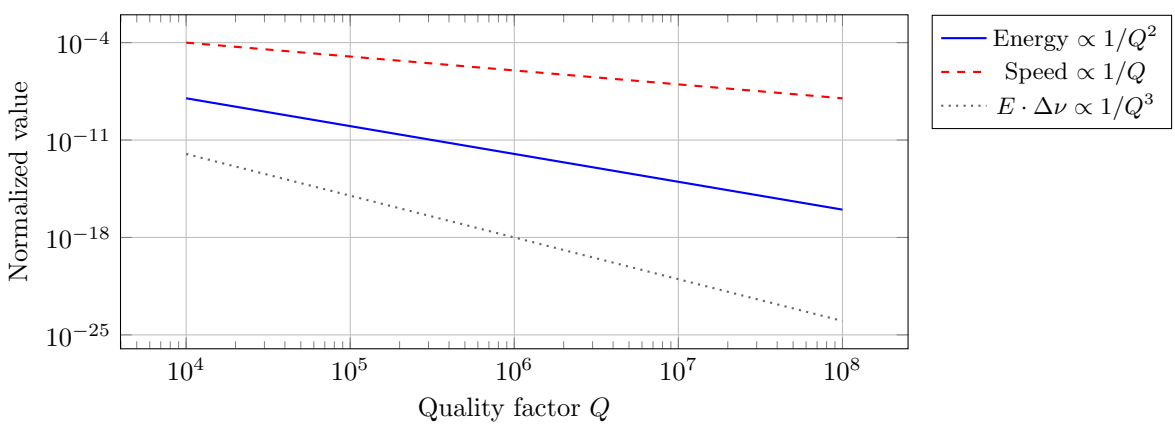
This product, sometimes called the energy-delay product or switching energy-bandwidth product, decreases only as $1/Q$ despite E decreasing as $1/Q^2$. The reason is that the savings in energy are partially offset by the lengthening of the photon lifetime: operating at higher Q concentrates energy more efficiently but accumulates it over a longer time window. The quantity $E \cdot \Delta\nu$ (energy times bandwidth) is therefore an intrinsic figure of merit of the resonant nonlinear device:

$$E \cdot \Delta\nu = \frac{n_g A_{\text{eff}}}{n_2 Q^2} \cdot \frac{c}{\lambda Q} \cdot \Delta\phi \propto \frac{1}{Q^3}. \tag{36}$$

While this still improves with Q, the point is that for a fixed target bandwidth B, the energy advantage from increasing Q is halted once $\Delta\nu = B$. Beyond that point, one is accumulating photon lifetime without improving bandwidth.

7.4 Speed Reduction: Physical Picture

The physical picture is illuminating: a high-Q resonator stores photons for a long time. This storage is precisely what enables the field to build up to high intracavity intensity and accumulate a large Kerr phase shift from a low input power. But this storage is also a memory: the resonator "remembers" what happened over a time τ_{ph} in the past. For rapidly time-varying inputs—such as the high-speed optical pulses in a neural network— this memory causes the neuron to temporally smear the input signal, degrading temporal resolution and limiting the maximum meaningful signal frequency to $\sim 1/\tau_{\text{ph}}$. In electronic terms, this is analogous to using a large capacitor to integrate charge efficiently, only to find that the RC time constant limits the circuit's frequency response.



8 Toward Energy-Efficient and High-Speed Photonic Neurons: Future Work

The analysis above establishes a fundamental tension: microring resonators with high Q provide dramatic energy savings ($\propto 1/Q^2$) but impose proportionally decreasing bandwidth ($\propto 1/Q$). For a photonic neuron operating at a target speed of, say, 10 Gbps, Eq. (32) restricts Q to at most $\sim 3 \times 10^4$, providing a relatively modest energy reduction of $\sim 10^3\times$ over the straight-waveguide baseline. Reaching femtojoule energies with gigahertz-scale bandwidth thus requires either circumventing this tradeoff through alternative structures or finding an operating point that satisfies both requirements simultaneously.

Future work will pursue two complementary approaches:

8.1 Alternative Structural Approaches

Coupled-resonator arrays. A photonic molecule consisting of two or more evanescently coupled rings can exhibit electromagnetically induced transparency (EIT)-like transmission profiles [2]. In such systems, a narrow dark-state resonance is nested within a broad bright-state envelope. The dark state can provide high-Q field enhancement for low-power operation, while the bright state's broad linewidth determines the overall coupling bandwidth. Careful engineering of the inter-ring coupling can, in principle, decouple the energy-storage Q from the input-coupling bandwidth.

Photonic crystal nanocavities. Photonic crystal (PhC) cavities achieve ultrahigh Kerr interaction by combining extreme mode volumes ($V_{\text{mode}} \sim (\lambda/n)^3$, orders of magnitude smaller than microring modes) with moderate $Q \sim 10^5$–10^6 [9]. Because the Kerr figure of merit scales as Q/V_{mode} (analogously to cavity QED cooperativity), high values can be reached at lower Q than a microring, thereby preserving more bandwidth for a given switching energy.

Pulsed resonant excitation. By matching the input pulse duration to τ_{ph} and exploiting the coherent field buildup during the pulse, one can achieve the full Kerr enhancement without committing to a persistent high-Q CW field. This approach trades the temporal averaging inherent in CW operation for a triggered, pulsed regime, and may allow GHz repetition rates with sub-ns pulses even for $Q \sim 10^5$–10^6.

8.2 Optimal Tradeoff Analysis

Rather than seeking to eliminate the Q–bandwidth tradeoff, a second strand of future work will perform a systematic optimization over the design space to identify the Pareto-optimal operating point for a given application. The figure of merit is the switching energy per neuron per second of computation, $E \cdot B_{\text{max}}^{-1}$, as a function of Q:

$$\frac{E}{B_{\text{max}}} \propto \frac{1}{Q^2} \cdot Q = \frac{1}{Q}. \tag{37}$$

This product decreases monotonically with Q, confirming that—for a given target bandwidth B—the optimal strategy is to operate at the highest Q consistent with $\Delta\nu \geq B$. In practice, thermal noise, fabrication uncertainty, and the onset of bistability introduce additional constraints. A constrained nonlinear optimization (e.g., via sequential quadratic programming) over the design variables $(R, g, T_{\text{operating}})$ and subject to the constraints $\Delta\nu \geq B_{\text{target}}$, $P_{\text{cav}} < P_{\text{bistability}}$, and $T_{\text{drift}} < \Delta\nu/(\partial\nu_r/\partial T)$ will yield a rigorous optimal design point for a specified neural network benchmark.

Numerical simulations using the finite-difference time-domain (FDTD) method and split-step Fourier solutions of the NLS equation (Eq. 3) will be used to validate the coupled-mode theory predictions, particularly in the nonlinear and near-bistable regimes where perturbative estimates break down.

9 Conclusion

High-Q microring resonators enable a $1/Q^2$ reduction in the input energy required to actuate the Kerr nonlinearity for optical neural switching. This scaling arises from the simultaneous resonant enhancement of intracavity intensity (one factor of Q from field buildup) and effective interaction length (one factor of Q from photon lifetime). Grounded in Agrawal's NLS framework and the SPM phase-shift formalism, we have shown that a well-designed Si_3N_4 microring with $Q = 10^6$–10^7 can operate in the femtojoule-per-operation regime at 1550 nm, consistent with the reported performance of state-of-the-art photonic neural hardware.

The same physics that confers this energy advantage—photon storage in a high-finesse cavity—also constrains the device bandwidth to $\Delta\nu = c/(\lambda Q) \sim 10$–$200$ MHz for the proposed quality factors. This restricts operation to well below the gigahertz rates desired for optical computing at scale. The energy-speed product $E \cdot \tau_{\text{ph}}$ scales only as $1/Q$ rather than $1/Q^2$, confirming that the energy gains are partially offset by lengthened response times.

Future work aims to break or circumvent this tradeoff through coupled-resonator designs, utilization of dispersion engineering to optimize microring resonator parameters, adiabatically tapered waveguides,

photonic crystal nanocavities and pulsed excitation schemes, or to identify the Pareto-optimal operating point through constrained system-level optimization. FDTD and split-step NLS simulations will provide quantitative validation prior to experimental fabrication.

References

[1] G. P. Agrawal, *Nonlinear Fiber Optics*, 5th ed. Academic Press, Waltham, MA, 2013.

[2] W. Bogaerts, P. De Heyn, T. Van Vaerenbergh, K. De Vos, S. K. Selvaraja, T. Claes, P. Dumon, P. Bienstman, D. Van Thourhout, and R. Baets, "Silicon microring resonators," *Laser & Photon. Rev.*, vol. 6, no. 1, pp. 47–73, 2012.

[3] J. S. Levy, A. Gondarenko, M. A. Foster, A. C. Turner-Foster, A. L. Gaeta, and M. Lipson, "CMOS-compatible multiple-wavelength oscillator for on-chip optical interconnects," *Nature Photon.*, vol. 4, pp. 37–40, 2010.

[4] M. H. P. Pfeiffer, J. Liu, A. S. Raja, T. Morais, B. Bhaskaran, and T. J. Kippenberg, "Ultra-smooth silicon nitride waveguides based on the Damascene reflow process: fabrication and loss origins," *Optica*, vol. 5, no. 7, pp. 884–892, 2018.

[5] X. Ji, F. A. S. Barbosa, S. P. Roberts, A. Dutt, J. Cardenas, Y. Okawachi, A. Bryant, A. L. Gaeta, and M. Lipson, "Ultra-low-loss on-chip resonators with sub-milliwatt parametric oscillation threshold," *Optica*, vol. 4, no. 6, pp. 619–624, 2017.

[6] P. P. Absil, J. V. Hryniewicz, B. E. Little, P. S. Cho, R. A. Wilson, L. G. Joneckis, and P.-T. Ho, "Wavelength conversion in GaAs micro-ring resonators," *Opt. Lett.*, vol. 25, no. 8, pp. 554–556, 2000.

[7] Y. Shen, N. C. Harris, S. Skirlo, M. Prabhu, T. Baehr-Jones, M. Hochberg, X. Sun, S. Zhao, H. Larochelle, D. Englund, and M. Soljačić, "Deep learning with coherent nanophotonic circuits," *Nature Photon.*, vol. 11, pp. 441–446, 2017.

[8] J. Feldmann, N. Youngblood, M. Karpov, H. Gehring, X. Li, M. Stappers, M. Le Gallo, X. Fu, A. Lukashchuk, A. S. Raja, J. Liu, C. D. Wright, A. Sebastian, T. J. Kippenberg, W. H. P. Pernice, and H. Bhaskaran, "Parallel convolutional processing using an integrated photonic tensor core," *Nature*, vol. 589, pp. 52–58, 2021.

[9] K. Nozaki, T. Tanabe, A. Shinya, S. Matsuo, T. Sato, H. Taniyama, and M. Notomi, "Sub-femtojoule all-optical switching using a photonic-crystal nanocavity," *Nature Photon.*, vol. 4, pp. 477–483, 2010.